Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/11/2005
PI Name:	Kanas, Nick M.D.		
Project Title:	Crew Member and Crew-Ground Interactions During International Space Station Missions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	415-750-2072
Organization Name:	University of California-San Francisco/Northern California Institute for Research & Education		
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City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-OLMSA-01
Start Date:	07/31/1998	**End Date:**	02/28/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Increment 2-9 Data collection complete. Awaiting final report at the end of the cooperative agreement.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marmar, Charles (University of California - San Francisco) Weiss, Daniel (University of California - San Francisco) Ritsher, Jennifer (University of California - San Francisco)		
Grant/Contract No.:	NCC9-0161		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During International Space Station (ISS) missions, international crews will be engaged in complicated activities over long periods of time. A number of interpersonal issues likely to impact on these missions must be addressed in order to ensure healthy crewmember interactions and optimal performance. A review of the literature of space analog studies on Earth, anecdotal reports from previous space missions, and the principal investigator's own work involving astronauts and cosmonauts have isolated crew tension, cohesion, and leadership role as important interpersonal issues. Cultural and language differences among crewmembers and the relationship of space crews with monitoring personnel in mission control also are important factors that influence crew performance and well-being. Hypotheses related to these interpersonal issues will be tested by having the crewmembers and personnel in ground control answer a series of questions from three standard mood and interpersonal group climate questionnaires, a critical incident log, and a culture and language questionnaire. The culture and language questionnaire will be administered once before each mission. The other measures will be completed on a weekly basis before, during, and after each mission and will take 15-20 minutes to fill out. The hypotheses will be tested using a piecewise linear regression model and the results will be discussed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These findings will have direct application for the development of countermeasures to reduce displacement and decrements in mood and interpersonal interactions during the International Space Station missions and even longer missions, such as an expedition to Mars. In addition, the interpersonal interactions of long-duration, multi-national space crews constitute a laboratory of small group behavior that tells us a great deal about ways in which groups of people on Earth can relate with a minimum of tension and improved cohesion when they are under stress. Finally, the ability of people from previously opposing political blocks to engage in complex activities, such as undertaking a space mission or conducting research together, serves as a model for international cooperation on Earth.
Task Progress:	This year we continued to collect data. Next year, we will compile and analyze the data.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Abstracts for Journals and Proceedings	Clarke L, Ritsher JB. "Humor as a potential training factor for International Space Station personnel." 75th Annual Scientific Meeting of the Aerospace Medical Association: Frontiers in Aerospace Medicine, Anchorage, Alaska, May 2-6, 2004. Aviation, Space, and Environmental Medicine, 2004 Apr;75(4 Suppl):B61. , Apr-2004
Abstracts for Journals and Proceedings	Kanas N, Ritsher JB, Ihle EC. "Positive effects of space travel. " 75th Annual Scientific Meeting of the Aerospace Medical Association: Frontiers in Aerospace Medicine, Anchorage, Alaska, May 2-6, 2004. Aviation, Space, and Environmental Medicine, 2004 Apr;75(4 Suppl):B28. , Apr-2004
Abstracts for Journals and Proceedings	Ritsher JB, Kanas N. "American-Russian cultural differences and the International Space Station." 75th Annual Scientific Meeting of the Aerospace Medical Association: Frontiers in Aerospace Medicine, Anchorage, Alaska, May 2-6, 2004. Aviation, Space, and Environmental Medicine, 2004 Apr;75(4 Suppl):B28. , Apr-2004
Articles in Peer-reviewed Journals	Kanas N. "Group interactions during space missions." Aviation, Space, and Environmental Medicine, 2004 Jul;75(7 Suppl):C3-5. PMID: 15267068 , Jul-2004
Awards	Kanas N, Manzey D. "International Academy of Astronautics Life Sciences Book Award, October 2004." Oct-2004
Presentation	Ritsher, J. B.; Ihle, E. C.; Kanas, N. "Positive psychological effects of space missions" 55th International Astronautical Congress, Vancouver, Canada, October 4-8, 2004. Paper #IAC-04-G.5.a.03. International Astronautical Federation, Paris, 2004. Oct-2004
Presentation	Clarke L.; Ritsher JB. "Soonoshenie mezhdu tipom lichnosti i chustvom yumora: Potentsialnoe primenenie dlya personala, rabotayushchego po programme MKS [Correlation between personality type and sense of humor: Potential applications for ISS program personnel]. " 4th International Congress of the Association of Aerospace, Marine, Extreme and Ecological Medicine of Russia. Moscow, 2004. Oct-2004
Presentation	Ritsher, J. B; Kanas, N.; Ihle, E. C.; Gushin, V. I. "Poleti v kosmos i ix positivnoe vliyanie na psihicheskoe sostoyanie [Space travel and positive mental health]. " 4th International Congress of the Association of Aerospace, Marine, Extreme and Ecological Medicine of Russia Moscow, 2004. Oct-2004